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$^{+4}$Notes of Complex Analysis

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** Abstrasct muito abstract . **

Notation ◇ Definition $(-)^{\diamond}$ Open $(-)^{\blacklozenge}$ Closed
□ Lemma ☐ Proposition ■ Theorem ⊠ Corollary

§ 1 Preliminaries

Let $\mathbb{C} := \mathbb{R}[i]$ the extension field of $\mathbb{R}$, obtained by adjoining i, the *imaginary unity*, which satisfies $i^2 = -1$. Some important details about $\mathbb{C}$:

- Any $z \in \mathbb{C}$ can be written as $z = x + iy$, where $x, y \in \mathbb{R}$ are the *real* and *imaginary parts* of z. Also, $\overline{z} := x - iy$ denotes the *complex conjugate* of z.
$$x = \mathrm{Re}(z) = \frac{z + \overline{z}}{2} \quad \text{and} \quad y = \mathrm{Im}(z) = \frac{z - \overline{z}}{2i}.$$
- $\mathbb{C} \simeq \mathbb{R}^2$, so there are natural geometric interpretations for the operations on complex numbers.
- The map $(z, w) \mapsto \langle z, w \rangle := z\overline{w}$ is a Hermitian inner product and induces the norm $|z|^2 = z\overline{z} = x^2 + y^2$ (the same as the usual for $\mathbb{R}^2$). Useful inequalities:
  i. $|z + w| \leq |z| + |w|$. ii. $\dfrac{1}{z} = \dfrac{\overline{z}}{|z|^2}$ (em $\mathbb{C}^\times$).
  iii. $||z| - |w|| \leq |z - w|$. iv. $|\mathrm{Re}(z)|, |\mathrm{Im}(z)| \leq |z|$.
- We can write $z \in \mathbb{C}^\times$ in *polar form* as $z = r\exp(i\theta) = r\,\mathrm{cis}(\theta)$, where $\theta = \arg(z)$ is the *argument* of z and $r = |z| > 0$ is the modulus.

Following the standard topology notation, given $z_0 \in \mathbb{C}$ and $r > 0$, $B(z_0; r) := \{z \in \mathbb{C} : |z - z_0| < r\}$, $B[z_0; r] := \overline{B(z_0; r)}$ and $\partial B(z_0; r) = \{z \in \mathbb{C} : |z - z_0| = r\}$.

$$\diamondsuit \quad \mathbb{D} := \{z \in \mathbb{C} : |z| < 1\}.$$

Naturally, $\mathbb{C}$ is a metric space and we work with the topology induced by the metric. In this context, open sets are completely characterized by the basis $\mathcal{B} := \{B(z_0; r) : z_0 \in \mathbb{C},\ r > 0\}$, along with all the associated facts and theory (compactness and connectedness in particular).

$$\diamondsuit \quad \operatorname{diam} \Omega := \sup_{\Omega} |z - w|.$$

Holomorphic functions

$\diamondsuit$ A function $f : \Omega^{\diamond} \subset \mathbb{C} \to \mathbb{C}$ is *holomorphic* at $z_0 \in \Omega$ if the limit,

$$\lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h}, \qquad !!\ h = h_1 + ih_2 \in \mathbb{C}$$

exists. When it exists, $f'(z_0)$ is called the *derivative* of f at z_0. We say holomorphic in Ω if $\forall z_0 \in \Omega,\ \exists f'(z_0)$, and *entire* if it is holomorphic on all $\mathbb{C}$.

Example Any polynomial function $p \in \mathbb{C}[z]$ is entire and $1/z$ is holomorphic on $\mathbb{C}^{\times}$. On the other hand, the conjugation $z \mapsto \overline{z}$ is not holomorphic. $\to$ Make the limit

Note. An equivalent and useless formulation for holomorphic at z_0 is that $\exists \alpha \in \mathbb{C} : f(z_0 + h) = f(z_0) + \alpha \cdot h + o(h)$, of course that $\alpha = f'(z_0)$.

$\square$ **2.2.** If f and g are holomorphic (in appropriate domains), then $f + g,\ fg,\ f/g$ and $g \circ f$ are holomorphic too. $\to$ Same formulas and proof of analysis on $\mathbb{R}^n$

$$F : \mathbb{R}^2 \to \mathbb{R}^2 \text{ diff } \not\Rightarrow F \sim f : \mathbb{C} \to \mathbb{C} \text{ homorphic}$$

Example $F : (x, y) \mapsto (x, -y) \sim f : z \mapsto \overline{z}$.

$$\diamondsuit \quad \partial_z := 1/2\,(\partial_x + 1/i\,\partial_y) \ \text{ and } \ \partial_{\overline{z}} := 1/2\,(\partial_x - 1/i\,\partial_y).$$

$\square$ **2.3.** If $f(z) = u(x, y) + iv(x, y)$ is holomorphic at $z_0 = x_0 + iy_0$, then

$$f_{\overline{z}}(z_0) = 0 \quad \text{and} \quad f'(z_0) = f_z(z_0) = 2u_z(z_0).$$

Also, $F(x, y) = (u(x, y), v(x, y))$ is diff (real sense) and $\det JF(x_0, y_0) = |f'(z_0)|^2$.

For first equal (Cauchy-Riemann equations) take the limits,

$$\lim_{h_1\to 0}\frac{f(z_0+h_1)}{h_1}=f'(z_0)=\lim_{ih_2\to 0}\frac{f(z_0+ih_2)}{ih_2}\Rightarrow\begin{cases}u_x=v_y\\u_y=-v_x\end{cases}.$$

Other affirmations are consequence of C-R equations,

$$JF(x_0,y_0)(h)=\begin{bmatrix}u_x & u_y\\-u_y & u_x\end{bmatrix}\begin{bmatrix}h_1\\h_2\end{bmatrix}=(u_x-iu_y)(h_1+ih_2)=f'(z_0)\cdot h,$$

also $\det JF(x_0,y_0)=u_x^2+u_y^2=|f'(z_0)|^2$.

■ **2.4.** Let $\Omega\simeq U^{\diamond}\subset\mathbb{R}^2$, $u,v\in C^1(U,\mathbb{R}^2)$ and $f=u+iv:\Omega\to\mathbb{C}$. If u and v satisfies the C-R equations, then f is holomorphic on Ω and $f'(z)=f_z$.

Write du and dv by extension and group them.

Power series

The most important examples are the exponential series, which converges absolutely on compact sets, and the geometric series, which converges absolutely on $\mathbb{D}$.

- $\exp(z)=\sum_{n=0}^{\infty}\frac{z^n}{n!}.\to\left|\sum\frac{z^n}{n!}\right|\le\sum\frac{|z|^n}{n!}=\exp|z|<\infty$
- $\frac{1}{1-z}=\sum_{n=0}^{\infty}z^n,\ |z|<1.\to S_N=\frac{1-z}{1-z}\sum^{N}z^n=\frac{1-z^{N+1}}{1-z}\to\frac{1}{1-z}$

In general, a *power series* is a sum of terms of the form $(a_nz^n)_{n\ge 0}$ and we interest in its absolute convergence.

■ **2.5.** Given a power series $\sum a_nz^n$, there exists $R\in[0,\infty]$ such that:

$$\sum_{n=0}^{\infty}|a_n||z|^n=\begin{cases}L, & |z|<R\\\infty, & |z|>R\end{cases}.$$

Moreover, if we use the convention that $1/0=\infty$ and $1/\infty=0$, then R is given by *Hadamard's formula*: $1/R=\limsup|a_n|^{1/n}$.

Let $L:=1/R$, for $|z|<R$ chooses $0<\epsilon\ll 1:(L+\epsilon)|z|<r<1$. Follows that $|a_n||z|^n\le\{(L+\epsilon)|z|\}^n=r^n$, geometric series. Similiar argument concludes divergence for case $|z|>R$.

The case $|z| = R$ is not conclusive

Example sdfjksjs

Euler's formulas for sine and cosine

- $\cos(z) = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!} = \frac{e^{iz} + e^{-iz}}{2}$
- $\sin(z) = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!} = \frac{e^{iz} - e^{-iz}}{2i}$

■ **2.6.** The power series $f(z) = \sum a_n z^n$ defines a holomorphic function in its disc of convergence. The derivative of f is also a power series,

$$f'(z) = \sum_{n=0}^{\infty} n a_n z^{n-1} =: g(z) \text{ (termwise differentiation)},$$

with same radius of converges as f.

Remember that $n^{1/n} \to 1$ as $n \to \infty$, so from Hadamard's formula follows the affimation of radius of convergence of g. The first assertion doesn't admite summaries, look if wants [1, Pág. 17]. Consists of an argument $\epsilon/3$, using,

$$f(z) = S_N(z) + E_N(z) = \sum_{n=0}^{N} a_n z^n + \sum_{N+1}^{\infty} a_n z^n.$$

⊠ **2.7.** A power series is an infinitely differentiable in its disc of convergence, and the higher derivatives are also a power series obtained by termwise differentiation.

Note. The same holds for series centered at $z_0 \in \mathbb{C}^\times$, it's simply translations.

Integration along curves

◇ A *smooth parametrized curve* is a function $z : [a, b] \to \mathbb{C}$ such that $\exists z'(t) \neq 0$ continuous on $[a, b]$, with lateral limits for $t = a, b$.

◇ Two parametrizations $z : [a, b] \to \mathbb{C}$ and $\widetilde{z} : [c, d] \to \mathbb{C}$ are *equivalent*, denote $z \sim \widetilde{z}$, if $\exists\, {}^1t : [a, b] \simeq [c, d]$ such that $t'(s) > 0$ and $\widetilde{z}(s) = z(t(s))$. The clases $[z]$ determines a *smooth curve* $\operatorname{Im} z =: \gamma \subset \mathbb{C}$ with orientation $z(a) \rightsquigarrow z(b)$.

Note. It's possible to extend this definitions naturally for *piecewise-smooth curve*. We have particular interest in *closed curves* $z \mid z(a) = z(b)$.

Example $\partial B(z_0; r) = z_0 + re^{it},\ t \in [0, 2\pi]$ (positive orientation).

$\diamondsuit$ Given a smooth curve $\gamma \subset C$ parametrized by $z : [a, b] \to \mathbb{C}$, and f a continuous function on γ, we define,

$$\int_\gamma f(z)\, dz := \int_a^b f(z(t))z'(t)\, dt.$$

Note. It's easy to see that the integral is independent of the parametrization. $\to$ Just use an equivalent parametrization and chain's rule

$$\diamondsuit \ \text{length}(\gamma) := \int_a^b |z'(t)|\, dt.$$

$\square$ **3.1.** Integration of continuous functions on curves satisfies:

i. $\displaystyle\int_\gamma f(z) + \alpha g(z)\, dz = \int_\gamma f(z)\, dz + \alpha \int_\gamma g(z)\, dz.$

ii. $\displaystyle\int_\gamma f(z)\, dz = -\int_{\gamma^-} f(z)\, dz.$ iii. $\displaystyle\left|\int_\gamma f(z)\, dz\right| \leq \sup_{z\in\gamma} |f(z)| \cdot \text{length}(\gamma).$

■ **3.2.** Let $F : \Omega^\diamond \subset \mathbb{C} \to \mathbb{C}$ holomorphic, $\Omega \supset \gamma : w_1 \rightsquigarrow w_2$ smooth curve and $dF(z) = F'(z) = f(z)\, dz$. Then,

$$\int_\gamma f(z)\, dz = \int_{\partial\gamma} F(z) = F(w_2) - F(w_1).$$

It follows by ■ (*Stokes*) or ■ (*TFC*) if you prefer.

⊠ **3.3.** In the setting of ■ 3.3. if γ is a closed curve, then, $\displaystyle\int_\gamma f(z)\, dz = 0.$

⊠ **3.4.** If f is holomorphic on $\Omega^\diamond \subset \mathbb{C}$ connected and $f' \equiv 0$, then f is constant. $\to$ It suffices to take a smooth curve $\gamma : w_0 \rightsquigarrow w$ and apply ■ 3.3.

§ 2 Cauchy's Theorem and Its Applications

References

[1] Stein, E. M., & Shakarchi, R. (2010). *Complex analysis (Vol. 2)*. Princeton University Press.